F78PA - Feedback on Student Performance - December 2012

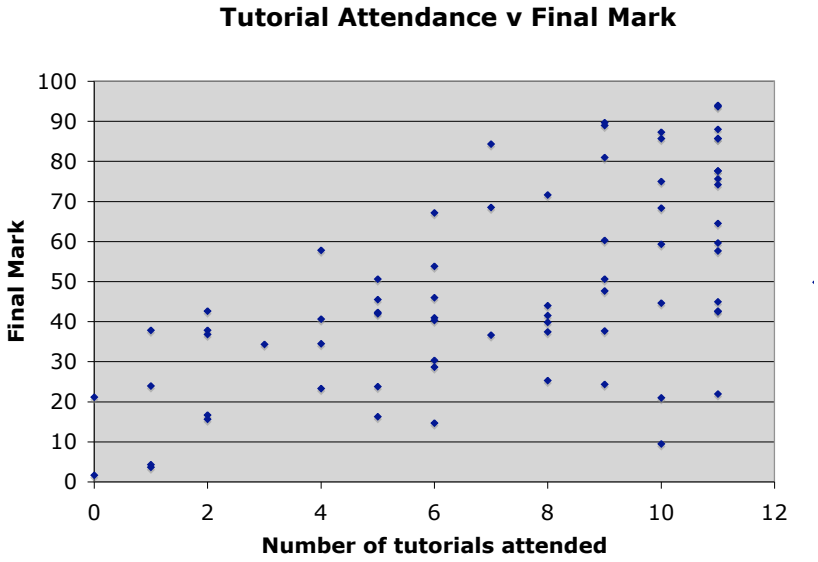
There were a number of very good final marks in this course (e.g. the highest final mark, 94, was obtained by two people and 12 students, out of 103 students, obtained a final mark of 80 or more), but overall student performance was disappointing. The pattern of performance is summarised in the table below:

A ($\geq$ 70)	19
B (60 - 69)	8
C (50 - 59)	9
D (40 - 49)	23
E (30 - 39)	16
F ($\leq$ 29)	24
Ave. mark	44.7
St. dev.	24.19

This pattern of performance is similar to the pattern of performance in past years, so some of the feedback given below is a repetition of feedback that was given last year and which you have already seen. Nevertheless, these points need to be re-iterated. **Please note:** The AMS department does not think that poor performance in this course is due to lack of student ability, but rather that there are other issues which some students need to address. In particular, it is recommended that students give careful consideration to the following issues:

Attendance

Student attendance, especially in tutorials, dropped off over the course of the term. Those students whose attendance was poor towards the end of the term missed out on some of the most important material in the course and they also missed the opportunity to get help on this material in the tutorial classes. The following plot shows the association between poor tutorial attendance and performance.



We can see from the plot above that that only a few students who attended 6 or fewer tutorial classes (out of 11 classes in total) obtained a D or better in the course, whereas most students who attended at least 8 tutorials obtained at least a D. It is clear from these data that a very simple strategy for improving performance is to improve attendance.

Work habits and learning strategies
Some students need to consider whether they have adopted work habits and learning strategies that are appropriate for university-level mathematics. The stem-leaf plot below summarises the student responses on the Course Questionnaires to the question: 'On average, how many hours per week did you spend on this module (not counting lectures and tutorials)?'

0	
1	x x x
2	x x x x
3	x x x x x x x
4	x x x x x x
5	x x x x
6	x x x
7	x x x
8	x
9	
10	x
11	x
12	

Note: The university recommends that students should work *at least* 6 hours per week outside class on *each course* that they are taking. We can see from the plot above that only 9 of the 32 respondents worked the recommended number of hours outside class on this course! These data, along with the disappointing student performance in this course, strongly suggest that another strategy for improving performance is to increase the time spent studying outside class!

In addition to increasing the time spent studying outside class, students should also consider whether their learning and studying strategies are *effective*. The university requires students to demonstrate that they *understand* the fundamental principles of their discipline and can *use* these principles to solve problems.
Important: This level of understanding and proficiency is *not* acquired by memorising formulas in the notes, or by learning the solutions to past papers and tutorial problems. Those strategies may have been useful at school but they will not lead to success at university.

Here are some *other* strategies that **do** lead to success at university:

- **Keep up to date with your lecture notes:** You need to work through your lecture notes as soon as possible after each lecture in order to gain a thorough understanding of the details in the notes (including all proofs!).
  Observation: Some students work through their notes by copying them out again, and this technique can be effective if you work out what is going on in the notes as you are copying them out. It is *not effective* to simply copy out things that you don't understand.

- **Make a *sustained* attempt to do the tutorial work *before* the tutorial class**: If you have worked through your lecture notes and have really understood them, you should be able to attempt the tutorial problems. The tutorial work gives you an opportunity to learn how to apply the ideas in the notes to solve problems. It takes time to solve problems and you must expect to devote time to your tutorial work.
  Note: Solving problems by copying an example in the lecture notes is not problem-solving - it is mimicry. Employers are looking for problem-solvers, not mimics.
  Finally: The more that you try to solve problems on your own, the better you will get at problem solving. You may even begin to enjoy working on hard problems - as Paul Erdos once said: 'A good problem is one that fights back'.

- **Learn from your mistakes and ask questions!** You can learn just as much from your mistakes as from doing a problem correctly! If you haven't solved a tutorial problem correctly, then you need to sort out where you went wrong and what concepts you haven't understood. If you get stuck trying to figure out where you went wrong out, then ask someone (friend or lecturer) for some help.
  Key point: If you haven't made a sustained effort to solve the tutorial problems or to figure out your lecture notes, you won't discover the weaknesses in your understanding.
  Important: You **cannot** learn how to solve problems or prepare for assessments by simply reading and memorising the solutions to tutorial problems. This is like learning to cook or to play a sport by watching someone do it on TV!

- **Do extra problems:** Once you have mastered the tutorial problems, you should try to find extra problems to solve. These can be found in the recommended textbooks which are usually available on short loan from the library. Past papers can also provide a useful source of practice problems, but do not fall into the trap of expecting your final assessment to be a re-tread of a past paper! It is a very dangerous strategy to put too much emphasis on learning the solutions to past papers.

- **Do not leave the work until the end of the term**. If you are only just beginning to sort out Tutorial Sheet 3 (for example) in Week 12, then you are simply too far behind in the course to do well on the final assessment.

- **Aim to master the entire course!** In general, many courses that you take at university build on ideas from other courses that you have taken. This means that your lecturers will expect you to have learned *everything* in the pre-requisite courses. So, if you skip a topic in a course because you have judged that you can pass the course without mastering that topic, you may be able to achieve the immediate short-term goal of obtaining a pass mark, but you are storing up trouble for yourself in future years.
  WARNING: If you did not obtain a final mark of **at least 50** in this course, then you should **plan to spend some time revising this material before the start of the third year**. Otherwise, **you will not be adequately prepared for Honours level courses next year**.

Conclusion: There are **no shortcuts** to deep understanding. The strategies recommended above are labour-intensive and students who adopt them will find that they are easily spending (at least) the 6 recommended hours per week working on each course! On the positive side, if you adopt these strategies, your problem-solving skills will improve and you will develop the work habits that are sought after in industry.

Mathematical proficiency
Finally, some students need to work to improve their mathematical skills. In particular, students are expected to be able to use calculus (differentiation and integration), manipulate inequalities, perform algebraic calculations, solve systems of linear equations, solve differential equations, solve quadratic equations, work withthe logarithm function, etc. *throughout* their degree course. If techniques or methods arise in your lecture notes and/or tutorial work and you don't remember how to use them, then you need to go back to the relevant mathematics lecture notes, revise the methods/techniques, and practice doing relevant calculations.

ADDITIONAL FEEDBACK on the FINAL EXAM

Detailed solutions to the exam questions can be found on Vision. Here are some additional comments about student performance on individual questions:

Question 1
The most common mistakes on this question were due to confusion about the definitions of and differences between (cumulative) distribution functions and density functions. The question asks you to determine whether then given functions are *density* function. **Note:** Density function *do not* have to be continuous! The only requirements are that

- $f(x) \geq 0$ for all $x \in \mathbb{R}$, and
- $\int_{-\infty}^{\infty} f(x)dx =$ Area under the graph of $f = 1$.

Also, a number of students didn't correctly compute the distribution function in part (c). For more details, see the model solutions.

Question 2

Most students could do part (a). The other two parts were not done very well by some students. In particular, in part (b) some students did not compute a *conditional* probability. See the model solutions for more details.

Question 3

(b) Some students did not have the correct limits of integration in the calculation of the marginal density $f_Y(y)$. Also, to specify the marginal density $f_Y(y)$, you must specify the value of the function for *every* $y \in \mathbb{R}$. If you did not make it clear that $f_Y(y) = 1 - \frac{y}{2}$ for $0 < y < 2$ and $f_Y(y) = 0$ *otherwise*, then 1 mark was deducted.

(c) This is another *conditional* probability calculation. The first step is to write:

$$\mathbf{P}(X > 1 \mid Y > 1) = \frac{\mathbf{P}(X > 1 \text{ and } Y > 1\}}{\mathbf{P}(Y > 1)}$$

The probabilities in the numerator and denominator can be computed using a geometrical argument (since the point (X, Y) is *uniformly* distributed in the triangle. Alternatively, the probability in the number can be computed by integrating the joint density over the correct region.

(d) Some students did not attempt this question and others attempted it but could not compute $\mathbf{P}(X+2Y \leq z)$ correctly because their limits of integration were wrong. See the model solutions for details.

Question 4

(a) One mark was deducted if you did not specify that $f_{X,Y}(x, y) = 2x^2e^{-xy}$ for $0 < x < 1, 0 < y < \infty$ and is equal to 0 everywhere else. Also, some students had trouble setting up the correct double integral of the joint density $f_{X,Y}(x, y)$ in order to compute $\mathbf{P}(Y > X)$. Often this was because they did not determine the region of integration correctly. Finally, some students could not determine the anti-derivative of the function $h(x) = 2xe^{-x^2}$. **Note:** You can find the anti-derivative of $h(x)$ by using the u-substitution $u = x^2$.

(b) To get full marks for this question, you had to identify that the *conditional density* for Y *given* that $X = x$ is an $Exp(x)$. Therefore, the conditional *expectation* $E(Y \mid X = x) = \frac{1}{x}$ since $\frac{1}{x}$ is the expected value for a $Exp(x)$ distribution.

Question 5

(a) This part was a bit of 'bookwork' . Students who had studied and understood the proof in the notes were able to do this part.

(b) This part was an application of part (a). Again, some students had difficulty with the conditional probability calculations and a number of students did not know how to compute a median. Calculation of the median of a distribution was covered on one of the tutorial sheets.

Question 6

(a) This part covered some of the theory about compound variables from the last part of the course. So, it you did not keep up with the notes and the tutorial, you would have had some difficulty doing this question.

(b) This was a standard application of compound variables.

Note: You will see compound variables and related idea in both the third year and the fourth year, so it is a good idea to make sure you really understand them.

Question 7

Many students were not able to state the Central Limit Theorem. This is a fundamental theorem and every student should be able to give the exact statement of this result and to apply it to approximate probabilities for sums of independent and identically distributed random variables. There were two applications of the Central Limit Theorem in this question and in both cases some students had problems applying the CLT - in particular, some students subtracted and divided by the wrong quantities.

Important: You will need to be able to use the Central Limit Theorem in almost all of your remaining courses - so learn what the theorem says and how to use it NOW.